Lesson 16: Further Aspects of Hypothesis Testing; Relationship between Testing and Interval Estimation

BSTA 551: Statistical Inference

Jessica Minnier

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Review: Where We Left Off

Lesson 15: Neyman-Pearson Lemma; Likelihood Ratio Tests

- The **Neyman-Pearson Lemma** shows the likelihood ratio test is most powerful for simple vs. simple hypotheses
- **UMP tests** exist for one-sided alternatives in exponential families (Karlin-Rubin Theorem)
- The **Likelihood Ratio Test (LRT)** $\Lambda = L(\hat{\theta}_0)/L(\hat{\theta}_{mle})$ provides a general recipe for composite hypotheses
- **Wilks' Theorem:** $-2\ln(\Lambda) \xrightarrow{d} \chi^2_m$ under H_0 (large-sample)
- Classic tests (z, t, F) are all LRTs

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Today's Goals:

1. Understand the distinction between **statistical significance** and **practical/clinical significance**
2. State and use the **duality theorem** connecting confidence intervals and hypothesis tests
3. Construct **confidence intervals by inverting tests** (the general method)
4. Understand **one-sided CIs** and their connection to one-sided tests
5. Use p-value duality and **equivalence testing** in medical research contexts
6. Understand **multiple testing** consequences for CI coverage

Roadmap

Topic	Key Idea	Textbook
Statistical vs. practical significance	A significant result may not be meaningful	Devore 9.6
Duality Theorem	$\theta_0 \in$ CI iff $H_0 : \theta = \theta_0$ is not rejected	Devore 9.6
Inverting a test	General method for constructing CIs	Devore 9.6
One-sided CIs	Linked to one-sided tests at level α	Devore 9.6
Equivalence testing	Testing that effects are *small enough*	Applied
Multiple testing & CIs	Simultaneous coverage	Devore 9.6

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Why This Matters

The duality between hypothesis tests and confidence intervals is one of the most elegant unifying results in statistics. It tells us that a confidence interval is **not just a range of plausible values** — it is a compact summary of all hypothesis tests we would fail to reject. In clinical trials, this connection is essential for both planning and reporting.

Statistical Significance vs. Practical Significance (Devore 9.6)

The Core Problem

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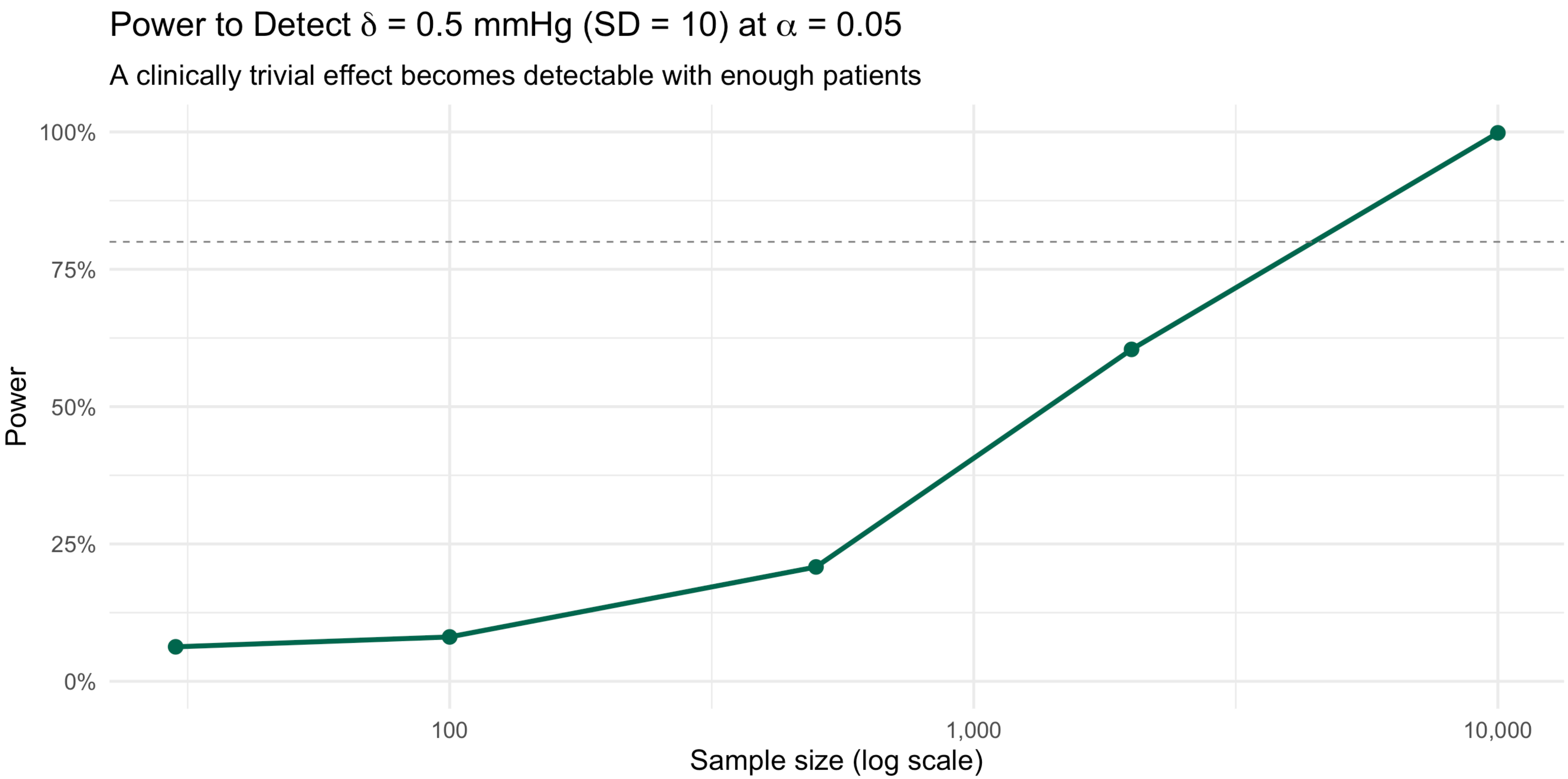
Medical Context

Suppose a new antihypertensive reduces systolic BP by 1 mmHg on average ($n = 50{,}000, p < 0.0001$). This is statistically significant – but is a 1 mmHg reduction clinically meaningful? The American Heart Association typically considers reductions of ≥ 5 mmHg clinically relevant.

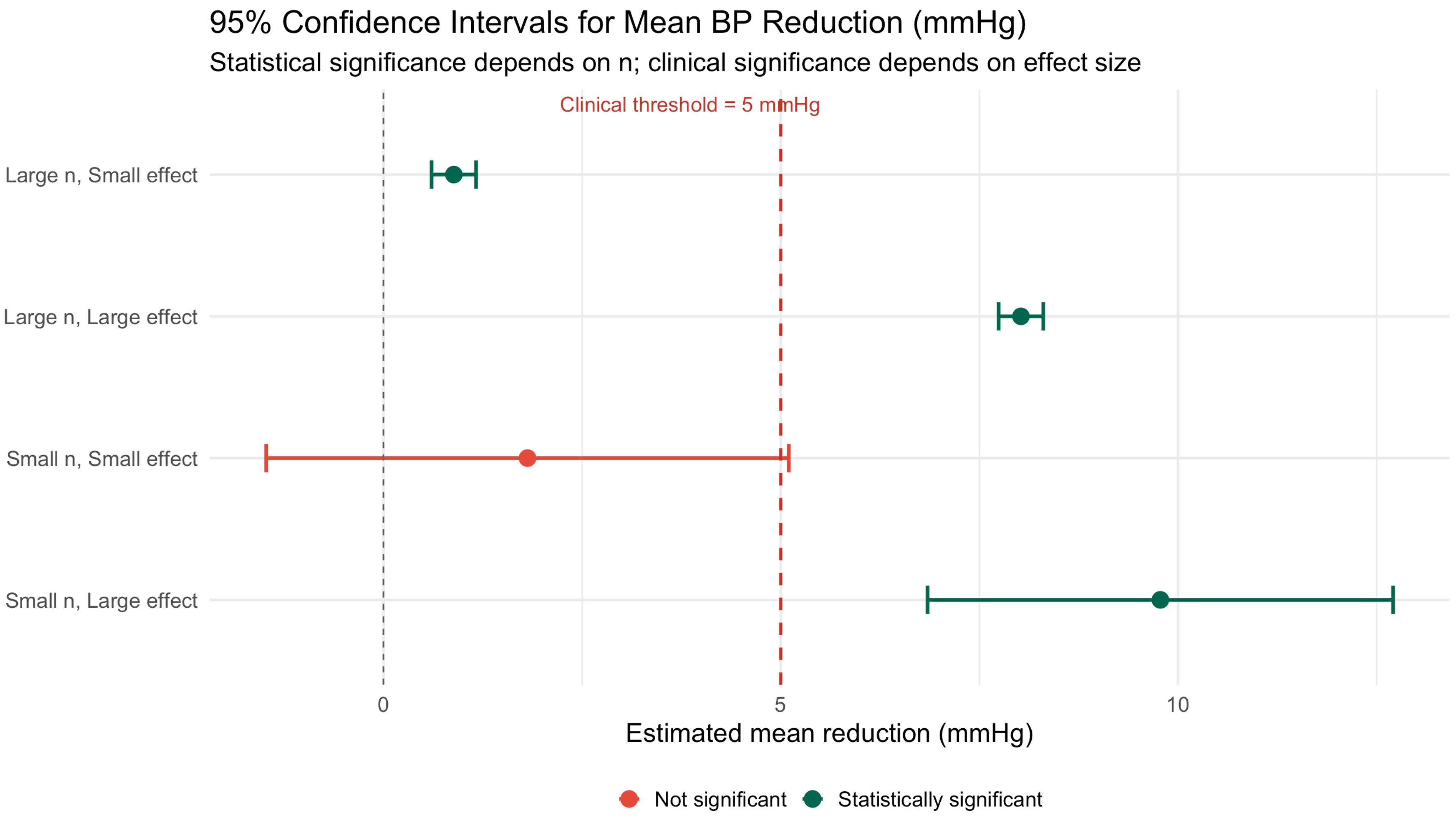
Large Sample Sizes Make Everything Significant

▶ Code



Visualizing Statistical vs. Clinical Significance

▶ Code



Reporting Standards: What Should We Report?

Best Practice (Devore 9.6 & Medical Journals)

Always report:

1. **The test statistic** (e.g., $t = 3.41$)
2. **The p-value** (e.g., $p = 0.001$)
3. **The confidence interval** for the effect size (e.g., 95% CI: [2.1, 8.7] mmHg)
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Why CIs are essential:

- A CI tells you **the range of plausible effects**, not just whether an effect exists
- A CI directly encodes practical significance: does the entire interval fall below the clinical threshold?
- A narrow CI near zero can rule out clinically meaningful effects even without rejecting H_0

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Example: CONSORT Guidelines

Major medical journals (NEJM, JAMA, Lancet) following CONSORT guidelines require authors to report effect sizes with CIs, not just p-values. The statement "$p < 0.05$" alone is considered insufficient.

Effect Size Measures

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Cohen's d (for means, σ known or estimated):

$$d = \frac{\mu_1 - \mu_2}{\sigma}$$

$\lvert d \rvert$	**Interpretation**
< 0.2	Negligible
$0.2 - 0.5$	Small
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Number Needed to Treat (NNT) for binary outcomes:

$$\text{NNT} = \frac{1}{p_{\text{control}} - p_{\text{treatment}}}$$

Interpretation: How many patients must be treated to prevent one additional event?

Example: Statin Therapy for MI Prevention

A landmark trial finds statins reduce the absolute risk of MI from 4.5% to 3.2% over 5 years.

$$\text{NNT} = \frac{1}{0.045 - 0.032} = \frac{1}{0.013} \approx 77$$

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Duality of Hypothesis Tests and Confidence Intervals

The Duality Theorem

Theorem: CI-Test Duality (Devore 9.6)

Let $C(\mathbf{X})$ be a $100(1-\alpha)\%$ confidence interval for a parameter θ, obtained by inverting a level-α test. Then:

$$\theta_0 \notin C(\mathbf{X}) \quad \Longleftrightarrow \quad \text{Reject } H_0 : \theta = \theta_0 \text{ at level } \alpha$$

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Intuition

A 95% CI for μ contains exactly those values of μ_0 for which the data are "consistent" (i.e., would not be rejected at $\alpha = 0.05$).

Proof of Duality

Consider testing $H_0 : \theta = \theta_0$ vs. $H_a : \theta \neq \theta_0$ using a two-sided level-α test.

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This is exactly the **95% confidence interval** $\bar{X} \pm z_{\alpha/2}\sigma/\sqrt{n}$!

So: θ_0 is in the CI $\iff$ we fail to reject $H_0 : \theta = \theta_0$. $\square$

The General Method: Inverting a Test

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Definition: Acceptance Region and Confidence Set (Devore 9.6)

For a level-α test of $H_0 : \theta = \theta_0$, the **acceptance region** $A(\theta_0) = R^C(\theta_0)$ consists of all sample outcomes that do NOT lead to rejection.

The **confidence set** is:

$$C(\mathbf{X}) = \{\theta_0 : \mathbf{X} \in A(\theta_0)\}$$

This set has coverage probability $\geq 1 - \alpha$ for all θ.

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Algorithm for inverting a test:

1. For each candidate value θ_0, determine the acceptance region $A(\theta_0)$
2. The CI is all values θ_0 for which $\mathbf{X} \in A(\theta_0)$

Example: Inverting the t-Test

Let $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with both μ, σ^2 unknown.

Test $H_0 : \mu = \mu_0$ vs. $H_a : \mu \neq \mu_0$ using $T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$.

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Acceptance region at level α:

$$A(\mu_0) = \left\{(x_1, \ldots, x_n) : -t_{\alpha/2,\, n-1} \leq \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \leq t_{\alpha/2,\, n-1}\right\}$$

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Invert: Find all μ_0 such that $\bar{x}$ falls in $A(\mu_0)$:

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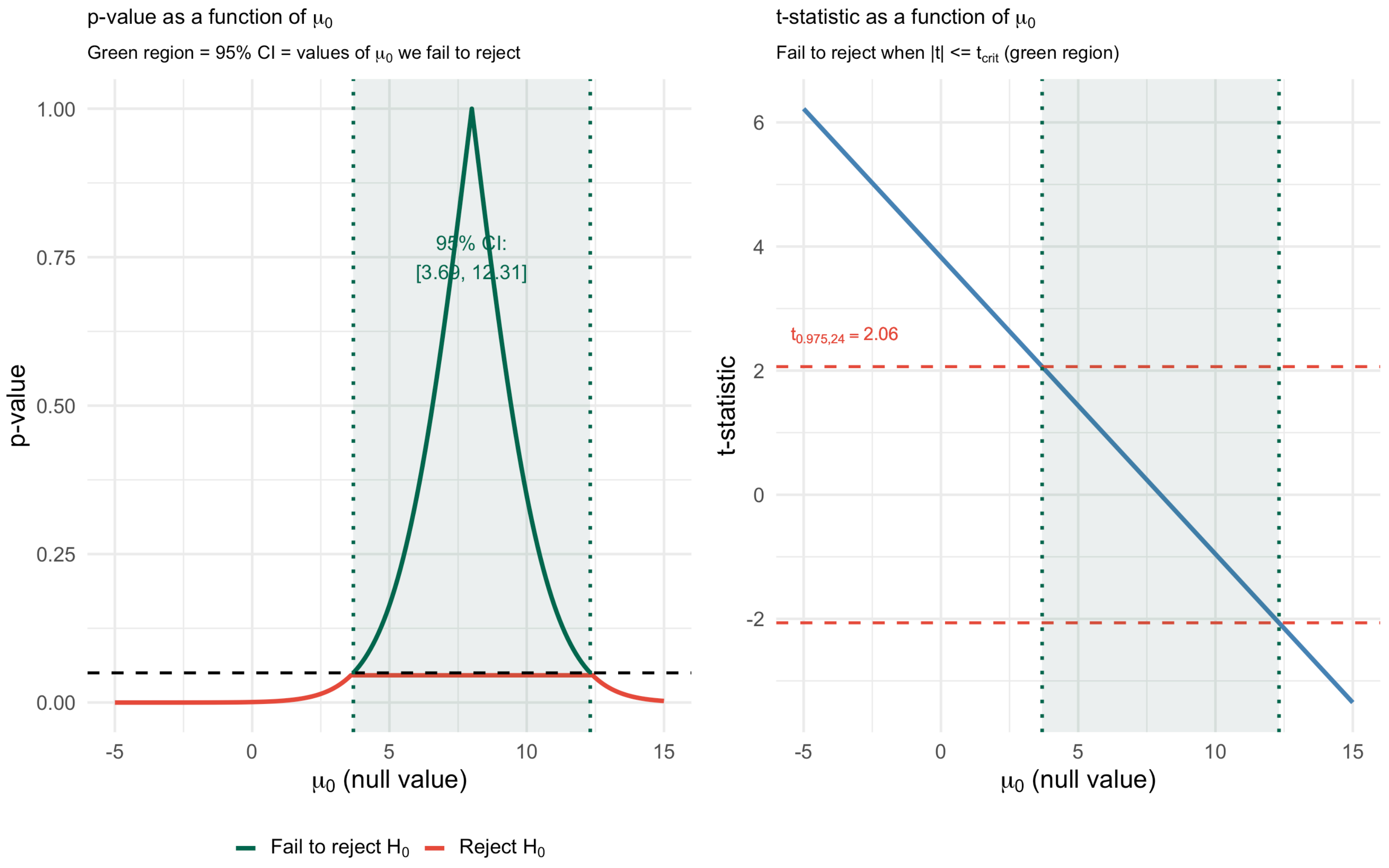
This recovers the familiar **one-sample t confidence interval.** The CI and t-test are **exactly dual** to each other.

Visualizing Duality: Clinical Trial Example

▶ Code



The p-Value Function

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$$p(\mu_0) = P_{H_0:\mu=\mu_0}\left(|T| \geq |t_{\text{obs}}|\right)$$

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Key properties:

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- Maximized at the MLE: $p(\bar{x}) = 1$
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Reading the p-Value Function

- The **width** of the region where $p(\mu_0) \geq 0.05$ equals the **width of the 95% CI**
- The **shape** tells us how evidence accumulates against each null value
- This function is more informative than a single p-value or a single CI

One-Sided Tests and One-Sided Confidence Intervals

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Theorem: One-Sided Duality (Devore 9.6)

Upper-tailed test: $H_0 : \theta \leq \theta_0$ vs $H_a : \theta > \theta_0$ at level α

Reject when $T > c_\alpha$

$\Leftrightarrow$ The corresponding $100(1-\alpha)\%$ **lower confidence bound** $\underline{\theta}$ satisfies:

$$\theta_0 < \underline{\theta}(\mathbf{X}) \quad \Longleftrightarrow \quad \text{Reject } H_0$$

Lower-tailed test: $H_0 : \theta \geq \theta_0$ vs $H_a : \theta < \theta_0$

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For the one-sample t-test:

Upper bound: $\bar{X} + t_{\alpha,n-1} \cdot S/\sqrt{n}$

Lower bound: $\bar{X} - t_{\alpha,n-1} \cdot S/\sqrt{n}$

Note: uses $t_{\alpha,n-1}$ (one-sided critical value), NOT $t_{\alpha/2,n-1}$

Summary: Two-Sided vs One-Sided Duality

Test Type	Reject H_0 when	Dual CI
Two-sided: $H_a : \theta \neq \theta_0$	$\|\hat{\theta} - \theta_0\|/SE > z_{\alpha/2}$	Two-sided: $\theta_0 \notin [\hat{\theta} \pm z_{\alpha/2} \cdot SE]$
Upper-tailed: $H_a : \theta > \theta_0$	$(\hat{\theta} - \theta_0)/SE > z_\alpha$	Lower bound: $\theta_0 < \hat{\theta} - z_\alpha \cdot SE$
Lower-tailed: $H_a : \theta < \theta_0$	$(\hat{\theta} - \theta_0)/SE < -z_\alpha$	Upper bound: $\theta_0 > \hat{\theta} + z_\alpha \cdot SE$

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Lower-tailed: $H_a : \theta < \theta_0$	$(\hat{\theta} - \theta_0)/SE < -z_\alpha$	Upper bound: $\theta_0 > \hat{\theta} + z_\alpha \cdot SE$

⚠ **Common Error**

A 95% **two-sided** CI uses $z_{0.025} = 1.96$, while a 95% **one-sided** bound uses $z_{0.05} = 1.645$. These are different! Don't use a two-sided CI to conduct a one-sided test at $\alpha = 0.05$.

Constructing CIs by Inverting Tests: The Pivot Method

The Pivot as the Bridge

The algebraic connection between a CI and a test often works through a **pivotal quantity** — a function of data and parameter whose distribution does not depend on the unknown parameter.

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Key pivots we have used:

Situation	Pivot	Distribution
μ, σ known	$Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$	$N(0,1)$
μ, σ unknown	$T = \frac{\bar{X}-\mu}{S/\sqrt{n}}$	t_{n-1}
σ^2	$\chi^2 = \frac{(n-1)S^2}{\sigma^2}$	χ^2_{n-1}
Large-sample LRT	$-2\ln\Lambda$	$\approx \chi^2_m$

Inverting the Chi-Squared Pivot for σ^2

Step 1: The pivot $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$.

Step 2: The acceptance region for testing $H_0 : \sigma^2 = \sigma_0^2$ is:

$$\chi^2_{1-\alpha/2,\, n-1} \leq \frac{(n-1)S^2}{\sigma_0^2} \leq \chi^2_{\alpha/2,\, n-1}$$

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Step 3: Solve for σ_0^2 to invert:

$$\frac{(n-1)S^2}{\chi^2_{\alpha/2,\, n-1}} \leq \sigma_0^2 \leq \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,\, n-1}}$$

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$$\chi^2_{1-\alpha/2,\, n-1} \le \frac{(n-1)S^2}{\sigma_0^2} \le \chi^2_{\alpha/2,\, n-1}$$

Step 3: Solve for σ_0^2 to invert:

$$\frac{(n-1)S^2}{\chi^2_{\alpha/2,\, n-1}} \le \sigma_0^2 \le \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,\, n-1}}$$

This is the **chi-squared CI for σ^2**, derived by inverting the chi-squared test.

Note: $\chi^2_{\alpha/2,n-1} > \chi^2_{1-\alpha/2,n-1}$, so dividing flips the inequality.

Example: Variability in Drug Concentration

In pharmacokinetics, it's important to control **variability** in drug plasma concentrations. A drug passes QC if $\sigma < 5$ ng/mL (so $\sigma^2 < 25$).

```r
# 15 measurements of drug plasma concentration (ng/mL)
set.seed(551)
conc <- rnorm(15, mean = 50, sd = 3.8)  # SD ≈ 3.8 (under QC limit)

n <- length(conc)
s2 <- var(conc)
df <- n - 1

# 95% two-sided CI for sigma^2
chi_upper <- qchisq(0.975, df)  # upper tail => lower bound of CI
chi_lower <- qchisq(0.025, df)  # lower tail => upper bound of CI

ci_var_low  <- df * s2 / chi_upper
ci_var_high <- df * s2 / chi_lower
ci_sd_low   <- sqrt(ci_var_low)
ci_sd_high  <- sqrt(ci_var_high)

cat("Sample variance:", round(s2, 3), "\n")
```

```
Sample variance: 17.82
```

```r
cat("Sample SD:", round(sqrt(s2), 3), "\n")
```

Inverting the LRT: Profile Likelihood CIs

For more complex models, we can invert the LRT using Wilks' theorem:

$$C(\mathbf{X}) = \left\{\theta : -2\ln\left(\frac{L(\theta)}{L(\hat{\theta})}\right) \leq \chi^2_{\alpha,1}\right\}$$

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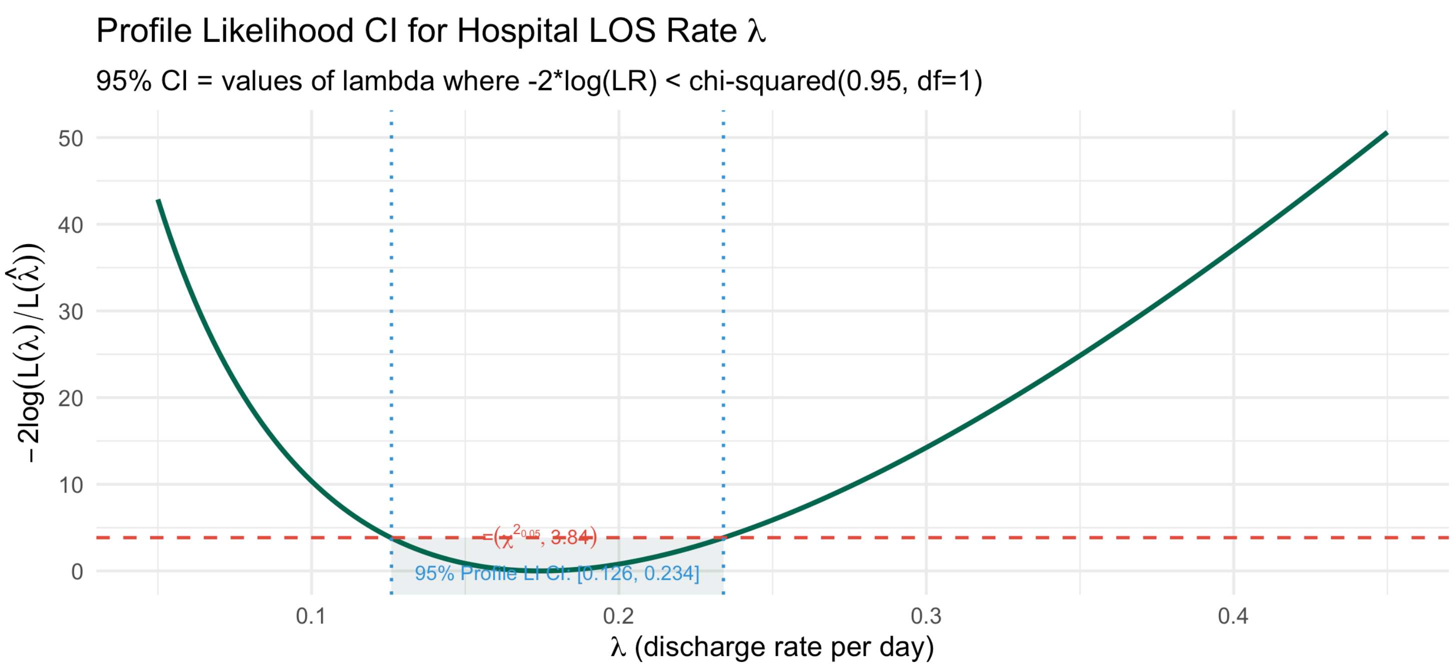
This is the **profile likelihood confidence interval**, valid for any one-dimensional parameter.

▶ Code



Multiple Testing and Confidence Intervals

Multiplicity: When We Test Many Hypotheses

When we conduct m tests simultaneously, even if all null hypotheses are true, we expect roughly $m \times \alpha$ false rejections by chance.

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By duality, if we construct m confidence intervals each with coverage $1 - \alpha$:

- Each individual CI has 95% coverage
- But the probability that **all m CIs simultaneously contain their true parameter** is less than 95%

$$P(\text{at least one CI fails}) = 1 - P(\text{all cover}) \leq 1 - (1 - \alpha)^m \approx m\alpha \text{ (Bonferroni)}$$

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Simultaneous Confidence Intervals

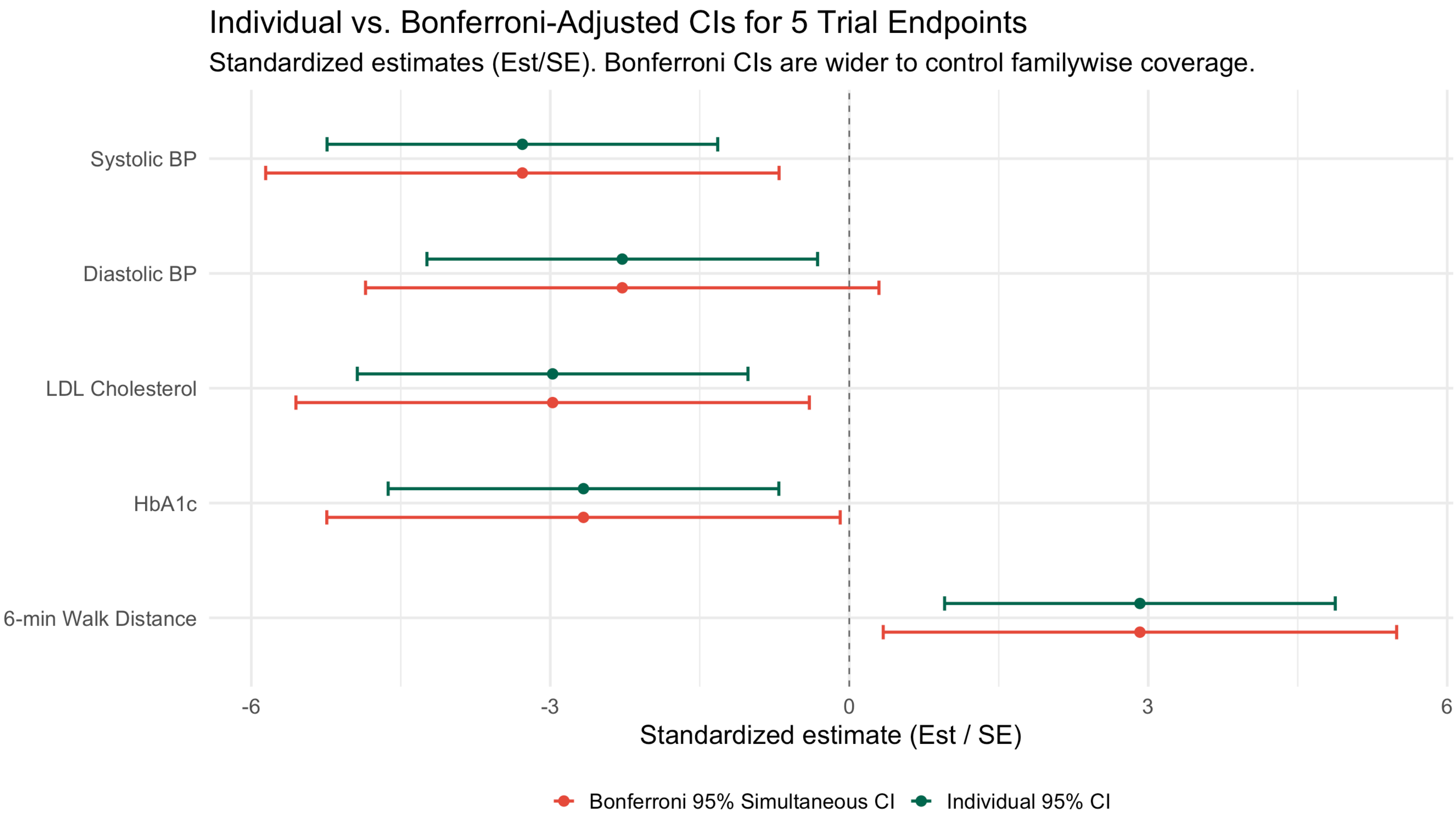
To ensure that **all m CIs simultaneously** contain their true values with probability $\geq 1 - \alpha$:

Use individual level $\alpha^* = \alpha / m$ for each CI (Bonferroni correction).

This produces **wider** intervals that account for multiplicity.

Example: Multiple Endpoints in a Clinical Trial

▶ Code



Devore 9.6 Textbook Examples

Devore Example 9.6: Testing Hypotheses About a Binomial Proportion

A clinical study examines whether a new screening test has sensitivity greater than 80%.

Let p = true sensitivity. We test $H_0 : p = 0.80$ vs $H_a : p > 0.80$.

In a sample of $n = 200$ subjects with disease, $X = 170$ test positive.

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In a sample of $n = 200$ subjects with disease, $X = 170$ test positive.

By exact binomial test and CI:

```r
# Exact binomial test
n_screen <- 200
x_pos    <- 170

binom_test <- binom.test(x_pos, n_screen, p = 0.80, alternative = "greater")
binom_test_two <- binom.test(x_pos, n_screen, p = 0.80, alternative = "two.sided")

cat("One-sided p-value:", round(binom_test$p.value, 5), "\n")
```

```
One-sided p-value: 0.04302
```

```r
cat("95% lower bound for p:", round(binom_test$conf.int[1], 4), "\n")
```

```
95% lower bound for p: 0.8021
```

```r
cat("\nTwo-sided 95% CI for p: [",
    round(binom_test_two$conf.int[1], 4), ",",
    round(binom_test_two$conf.int[2], 4), "]\n")
```

```
Two-sided 95% CI for p: [ 0.7928 , 0.8965 ]
```

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CI-Test Duality: Sweeping Over Null Values

We already computed a 95% CI for mean BP reduction from the blood pressure data: $[\hat{\mu} - t_{0.025,24} \cdot s/\sqrt{n},\ \hat{\mu} + t_{0.025,24} \cdot s/\sqrt{n}]$.

The duality theorem says this interval is **exactly the set of null values we would fail to reject**. Let's verify this directly by testing several specific values of μ_0.

▶ Code

95% CI: [3.69, 12.31]. xbar = 8.00, s = 10.45, n = 25

mu_0	t-statistic	p-value	Decision	In 95% CI?	Duality holds?
0.0	3.828	0.001	Reject H_0	✗ No	✗ No
2.0	2.871	0.008	Reject H_0	✗ No	✗ No
3.7	2.057	0.051	Fail to reject H_0	✓ Yes	✗ No
5.0	1.435	0.164	Fail to reject H_0	✓ Yes	✗ No
8.0	0.000	1.000	Fail to reject H_0	✓ Yes	✗ No
12.3	-2.057	0.051	Fail to reject H_0	✓ Yes	✗ No
10.0	-0.957	0.348	Fail to reject H_0	✓ Yes	✗ No
12.0	-1.914	0.068	Fail to reject H_0	✓ Yes	✗ No

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Key Takeaways and Looking Ahead

Key Takeaways

Today's Main Points

1. **Statistical significance $\neq$ practical/clinical significance**: always report effect sizes and CIs alongside p-values
2. **CI-Test Duality (Devore 9.6):** $\theta_0 \in$ CI(X) $\iff$ fail to reject $H_0 : \theta = \theta_0$ at the same α
3. **Inverting a test** provides a general, principled method for constructing CIs from any hypothesis test
4. **One-sided duality:** a $100(1-\alpha)\%$ one-sided bound corresponds to a one-sided test at level α (using z_α, not $z_{\alpha/2}$)
5. **Multiple testing:** simultaneous CIs must be widened (e.g., Bonferroni) to maintain joint coverage
6. **Profile likelihood CIs** invert the LRT using Wilks' theorem: a general tool for any parameter

Looking Ahead

Next class (Lesson 17): Bootstrap Hypothesis Testing (C&H Chapter 8)

- The bootstrap hypothesis test: resampling under H_0
- One-sample and two-sample bootstrap tests
- Permutation tests: exact null distribution by permuting labels
- Comparison: parametric vs. bootstrap vs. permutation tests
- Applications to clinical trial data

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Section 9.6.
- Chihara and Hesterberg. *Mathematical Statistics with Resampling and R* (Wiley). Chapter 8.
- Wasserstein, R. L., & Lazar, N. A. (2016). The ASA statement on p-values: context, process, and purpose. *The American Statistician*, 70(2), 129–133.

Questions?

Thank you!